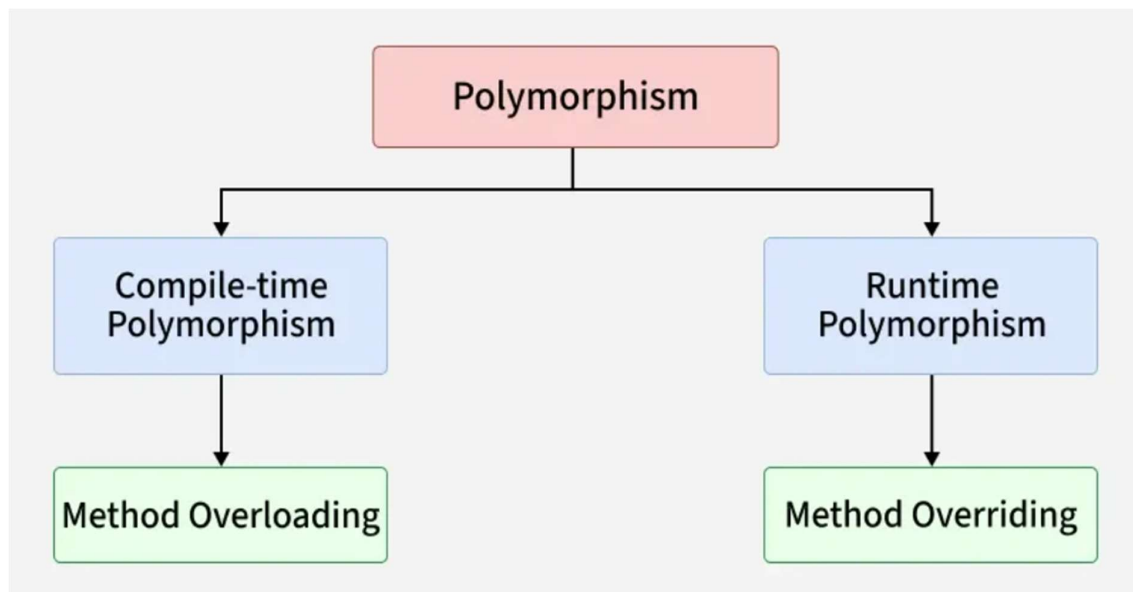
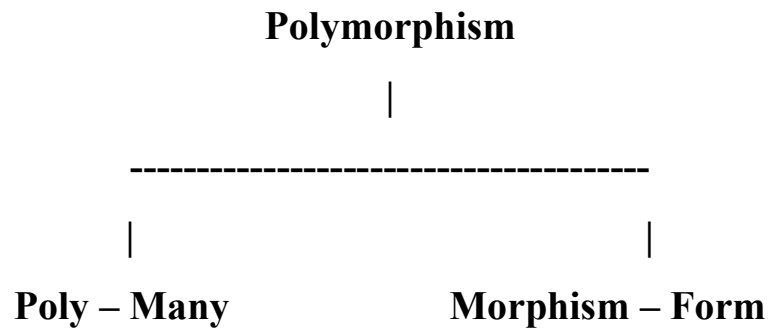


Polymorphism in Java

What is Polymorphism :-

- Polymorphism in Java is a **core concept of Object-Oriented Programming (OOP)** that allows objects to take on "many forms."
- It enables a single interface to be used for multiple underlying **data types or implementations, promoting flexibility, code reusability, and scalability.**



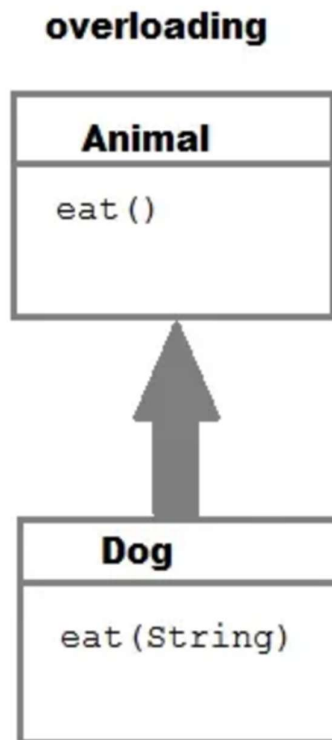
Method Overloading

Method Overloading :-

- Method overloading in Java is also known as **Compile-time Polymorphism** and **Static Polymorphism**.
-

Different Ways of Method Overloading :-

- **No Parameter**
- **By Passing Parameter**
- **By Passing number of parameter**
- **By changing Sequence** of parameter
- **By changing the datatype**



Implementation of Method Overloading

Input :-

```
1 package Oops;
2
3 public class Methodoverloading {
4
5     void payment() //by passing no parameter
6     {
7         System.out.println("Ccredit");
8     }
9
10    void payment(int num) //by passing parameter
11    {
12        System.out.println("Debit");
13    }
14
15    void payment(int r,String name) //by passing number of parameter
16    {
17        System.out.println("Cash on Delivery");
18    }
19
20    void payment(String p, int c) //by passing the sequence of parameter
21    {
22        System.out.println("UPI");
23    }
24
25    void payment(float z,char w) //by passing the datatype
26    {
27        System.out.println("Paytm");
28    }
29
30    public static void main(String[] args) {
31
32        Methodoverloading ob=new Methodoverloading();
33        ob.payment();
34        ob.payment(100);
35        ob.payment(20, "Cravita");
36        ob.payment("Nikhil", 10);
37        ob.payment(1.0f, 'A');
38    }
39 }
```

Output :-

```
Ccredit
Debit
Cash on Delivery
UPI
Paytm
```

Input :-

```
1 package Oops;
2 public class Methodoverloading_Example {
3
4     void area(int sum)
5     {
6         System.out.println("Area of Sqaure is : " +(sum*sum));
7     }
8
9     void area(int length, int breadth)
10    {
11        System.out.println("Area of Rectangle is : " +(length*breadth));
12    }
13
14    void area(float r)
15    {
16        System.out.println("Area of Circle is : " +(3.14*(r*r)));
17    }
18
19    void area(float b, int h)
20    {
21        System.out.println("Area of Triangle is : "+(1/2)* (b*h));
22    }
23    public static void main(String[] args) {
24
25        Methodoverloading_Example ob = new Methodoverloading_Example();
26        ob.area(6);
27        ob.area(4, 8);
28        ob.area(3f);
29        ob.area(5.2f, 4);
30    }
31 }
```

Output :-

```
Area of Sqaure is : 36
Area of Rectangle is : 32
Area of Circle is : 28.26
Area of Triangle is : 0.0
```